

Minor / Specialization in Quantum Technologies

Minor Details

Minor name	Minor / Specialization in Quantum Technologies
Eligibility batch(es)	Batch 2024 onwards

Table 1: Minor / Specialization in Quantum Technologies — Requirements

Sl.	Course code	Course name	L-T-P-C	Credits
QT-301 and QT-302 are both mandatory QT-301 and one of QT-302 / PH-513 / EP-301 are required.				
1	QT-301	Survey of Quantum Technologies and Applications	3-0-0-3	3
2	QT-302 / PH-513 / EP-301	Foundations of Quantum Technologies	3-0-0-3	3
At least one of QT-303P and QS-501P / QT-304P is mandatory Choose one laboratory course.				
3	QT-303P	Basic Programming Lab	2-0-1-3	3
4	QS-501P / QT-304P	Experiments in Quantum Optics / Basic Laboratory Course for Quantum Technologies	2-0-1-3	3
At least one of QT-405, QT-406, QT-407 and QT-408 is mandatory Choose one course from this basket.				
5	QT-405 / CS-520	Introduction to Quantum Computation	3-0-0-3	3
6	QT-406	Introduction to Quantum Communication	3-0-0-3	3
7	QT-407	Introduction to Quantum Sensing	3-0-0-3	3
8	QT-408 / PH-532 / PH-501	Introduction to Quantum Materials	3-0-0-3	3
Optional / Additional Courses Optional for the minor. For specialization, complete enough additional courses from this basket to reach at least 18 QT credits.				
9	QT-509	Engineering Foundations of Quantum Technologies	3-0-0-3	3
10	QT-510 / PH-601	Solid State Physics for Quantum Technologies	3-0-0-3	3
11	QT-511 / PH-550	Introduction to Quantum Optics	3-0-0-3	3
12	CS-521	Introduction to Post Quantum Security	3-0-0-3	3